



Project Report

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	25
Understanding/Analysis	7					
Implementation	8					
Report Writing	10					
Total obtained mark						
Comments						

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Course Name: System Analysis and Design Capstone Project

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Project Proposal: Psychological Support System

Project Title: Psychological Support System (PSS)

1.Introduction

Mental health is a key aspect of our overall well-being, yet many individuals struggle to locate the correct help when they need it the most. The Psychological Support System (PSS) is a conceptual initiative that aims to offer a secure, accessible, and organised environment in which people can get mental health treatment. This digital platform intends to provide AI-powered initial evaluations, professional counselling directories, peer support networks, and self-help tools to assist users in navigating their mental health journey. This project is being created as part of a System Analysis and Design Capstone Project, with a significant emphasis on system documentation, workflow design, and functional visualisation - rather than real implementation or coding. Our mission is to knock down challenges to mental health concern and build a system that actually fulfils the needs of individuals, whether they are students, employees, or anybody dealing with emotional difficulties.

2.Problem Statement

Plenty of people desire to get assistance with their mental health, yet they face difficulties like:

- **Limited Accessibility:** Finding a professional therapist or counselor can be difficult and expensive.
- **Stigma & Privacy Concerns:** Many fear being judged for seeking mental health support.
- **Lack of Peer Support:** The absence of a strong support network can make struggles feel isolating.

- **Resource Constraints:** There is a shortage of mental health professionals, leading to long wait times for therapy sessions.

The PSS seeks to close these gaps by offering an automated, centralised support system that includes AI-powered assistance, professional counselling links, peer groups, and self-help materials.

3.Objectives

The Psychological Support System seeks to:

- Provide chatbot assistance driven by AI to offer preliminary mental health evaluations and counselling.
- To provide expert assistance, compile a list of certified counsellors.
- Create a forum for peer support where people can interact and exchange stories.
- Offer self-help resources like journaling activities and meditation manuals.
- Include resources for emergencies, such as hotlines and ways to get help right away.

4.Scope

The PSS will prioritise the following major areas:

- **AI Chatbot Support:** A virtual assistant that helps users identify their mental health needs.
- **Professional Counselor Directory:** A structured list of certified professionals, with appointment booking options.

- **Peer Support Forum:** A safe space for discussions and emotional support from others who understand.
- **Self-Help Resources:** A collection of mindfulness exercises, therapy guides, and educational content.
- **Emergency Contact Integration:** Direct access to crisis helplines for urgent support.

5.Methodology

This project uses a structured system design method, which includes:

5.1 Requirement Analysis

- Perform enquiries with mental health professionals and potential consumers to gain perspectives.
- Identify the system's primary functionality and restrictions.
- Prepare a requirement specification document covering features and objectives.

5.2 System Design

- Create system architecture and workflow diagrams for individual modules.
- Create prototypes for user interfaces, including as dashboards, counsellor portals, and chatbots.

5.3 Conceptual Testing

- Validate system design using real-world situations and use cases.
- Refine the design to address possible issues.

5.4 Documentation

- Create a detailed project report including system design, diagrams, and processes.
- Provide user manuals and recommendations for future adoption.

6. Diagrams

The system design will be represented by the following diagrams:

6.1 Use Case Diagram

- **Actors:** User, Psychologist, Psychiatrist, Admin.
- **Use Cases:** User Registration & Login, Self-Assessment, Book an Appointment, Attend a Counseling Session, Chat with AI Support, View Mental Health Resources, Post in the Forum, Track Therapy Progress, Set Mental Wellness Goals, Connect with a Support Buddy, Daily Mood Check-In, Pay for Therapy Session, Rate & Review Therapist, Receive Session & Activity Reminders, View & Manage Patient Profiles, SOS Button Activation, View Progress Report
- **Purpose:** Show interactions between actors and the system.

6.2 Data Flow Diagram (DFD)

- **Level 0 (Context Diagram):** High-level overview of data flow.
- **Level 1:** Breakdown of chatbot assessment, counseling sessions, and forum interactions.
- **Purpose:** Illustrate how data moves through the system.

6.3 Activity Diagram

- **Processes:** Chatbot conversation flow, counselor booking, self-help resource usage.
- **Purpose:** Represent the flow of activities and decision points in the system.

6.4 State Diagram

- **States:** Chatbot states (Initial Query, Assessment Complete), Booking states (Pending, Confirmed, Completed).
- **Transitions:** Events that trigger state changes.
- **Purpose:** Show the lifecycle of system interactions.

6.5 Sequence Diagram

- **Interactions:** User interacts with chatbot → Chatbot provides response → User books a counselor session.
- **Purpose:** Depict the sequence of interactions between users and the system.

6.6 Class Diagram

- **Classes:** User, Patient, Therapist, Admin, Appointment, Moodlog, CounselingSession, Chatbot, Resource, Forum, Post, EmergencyContact, Security, Report, Payment, Notification.
- **Purpose:** Represent the system's structure using classes and relationships.

6.7 Entity-Relationship (ER) Diagram

- **Entities:** User, Counselor, Chatbot, SupportForum, Booking.
- **Relationships:** A user interacts with a chatbot, a user books a counselor, a user participates in forums.
- **Purpose:** Define the database structure and entity relationships.

7.Expected Outcomes

- A well-documented system design for the Psychological Support System.
- Easier access to mental health resources for users who need support.
- Reduced stigma through AI-driven, anonymous assessments.
- A comprehensive project report with detailed system design diagrams.

8. Timeline

Phase	Duration	Deliverables
Requirement Analysis	1 week	Requirement specification document
System Design	2 weeks	Architecture and diagrams
Conceptual Testing	1 week	Real-world scenario validation
Documentation	1 week	Final project report

9. Team Members & Roles

Role	Responsibilities	Assigned
Project Manager	Oversees planning & coordination.	Farhan Ibn Mustafa
System Analyst	Designs system architecture & workflows.	Md. Mobin
Documentation Specialist	Prepares final reports and user guides.	Yeasmin Akter

10. Conclusion

The Psychological Support System (PSS) is intended to offer a human-centered, supportive, and affordable system for mental health care. By focussing on organised system design and documentation, we are creating a road map for future deployment that will significantly improve how individuals seek and get mental health care. Our objective is to build a support system that is scalable, stigma-free, and inclusive, promoting emotional well-being for everyone.